

How Smartphones, Cloud and Data Analytics are Driving RF Test



Abhishek A. Mutha
is a senior technical
correspondent at EFM

To succeed in dealing with the most demanding radio frequency (RF) design challenges, engineers need instruments with the latest technology and capabilities. Here, we take a look at some interesting test equipment for RF applications that have been recently introduced in the market. We also highlight how these tools are enabling engineers solve problems in designs and maintain existing RF systems and infrastructure.

Wider real-time bandwidth driving better designs in defence

Spectrum analysers have been used for the development of electronic warfare and radar systems for a long time now. With traditional measurements becoming inadequate for today's modern systems, a signal analyser with real-time spectrum analysis and vector signal analysis could be an alternative for spectrum analysers.

Vishal Gupta, senior application consultant (RF/MW, Surveillance), Keysight Technologies, says, "When you look at platforms to perform wide-band and radar-signal analysis, earlier the widest real-time bandwidth available was 160MHz. Now, there is up

to 510MHz of analysis bandwidth available."

The wide bandwidth range up to 26.5GHz permits accurate measurement of parameters like wideband chirp linearity in advanced radar systems. Engineers can use the intuitive, multi-touch 35.8cm (14.1-inch) display to pinch or zoom signals for better analysis. Based on proprietary technologies, it allows for a deeper analysis of transient, wideband and elusive signals.

Gupta says, "There has always been demand for higher analysis and real-time bandwidth, mainly from aerospace and defence engineers who want to analyse their wide-band radar and electronic warfare and satellite signals."

He adds, "The introduction of UXG series signal analyser with real-time spectrum capability will bridge this gap."

Anticipating and fixing problems before these fail your radio system

From police personnel and fire departments to usage in private sector activities such as construction, security and maintenance, land mobile radio (LMR) systems have been extensively adopted for communication between geographically-dispersed teams and mobile personnel over pre-defined frequencies.

Today, LMR is also used by small- and medium-sized organisations to consolidate their business operations. The market is slowly witnessing a growing approval for digital LMR systems and it is also anticipated that its adoption will surpass analogue LMR in the years to come.

An LMR system can be as simple as a base station and two handheld units or as complex as hundreds of mobile units and various other devices. At any level, it is important to maintain and monitor the health of an LMR system and its components.



Fig. 1: N9040B UXG X-Series signal analyser
(Image courtesy: www.literature.cdn.keysight.com)



Fig. 2: Model 3141 channel power monitor